

Cambridge (CIE) IGCSE Physics
Paper 3: Theory (Set C)
Core

Saturday 26 April 2025

Morning (Time: 1 hour 15 minutes)

Total marks

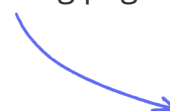
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Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8N (acceleration of free fall = 9.8m/s^2).
- You should show all your working and use appropriate units.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



- 1 (a)** A student has a pile of A4 paper for his computer printer.

Fig. 1.1 shows the dimensions of the pile of paper.

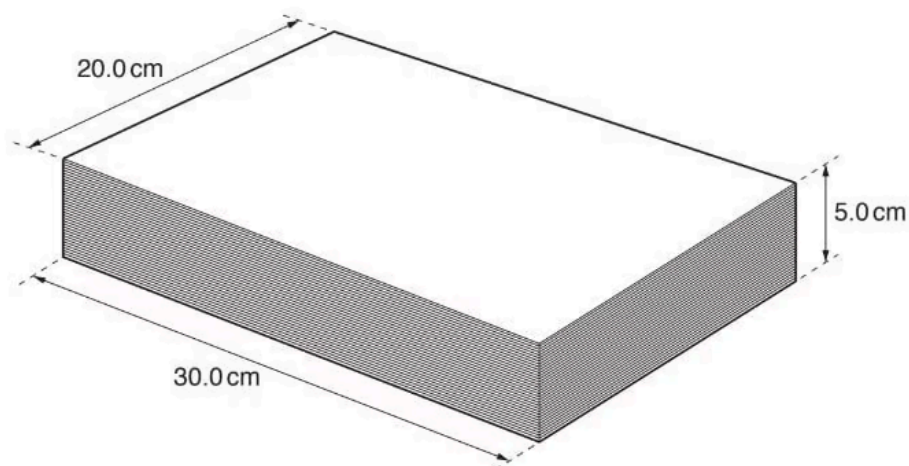


Fig. 1.1 (not to scale)

The pile contains 500 sheets of paper.

Calculate the average thickness of one sheet of paper.

average thickness = cm

(1 mark)

- (b)** Show that the pile of paper has a volume of 3000 cm^3 . Use the information shown in Fig.1.1.

(1 mark)

- (c)** The student measures the total mass of the paper in the pile.

State the name of a device used to measure mass.

(1 mark)

(d) The mass of the paper in the pile is 2400 g.

Calculate the density of the paper.

density = g/cm³

.....

.....

.....

(3 marks)

2 (a) Fig. 2.1 shows students getting onto a school bus.



Fig. 2.1

A student describes part of the journey.

The bus accelerates from rest at a constant rate for 10 s. It reaches a maximum speed of 10 m/s.

The bus maintains a constant speed of 10 m/s for 60 s.

The bus then decelerates at a constant rate for 15 s, until it stops.

On Fig. 2.2, draw the speed-time graph for this part of the journey made by the bus.

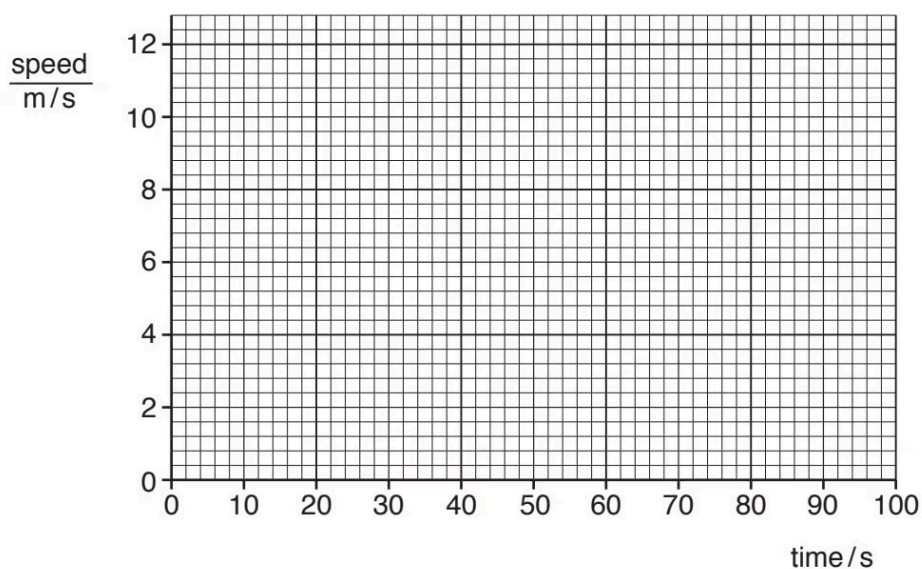


Fig. 2.2

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.....

.....

.....

(5 marks)

(b) On another part of the journey, the average speed of the bus is 7.5 m/s.

Calculate the distance the bus travels in 150 s.

distance = m

.....

.....

.....

(3 marks)

3 (a) Explain, in terms of their molecules, why the density of water is greater than that of oil.

.....
(1 mark)

(b) State the difference between mass and weight.

.....
(1 mark)

(c) The mass of a bucket is 1.2 kg.

When the bucket is filled with oil the total mass is found to be 14.1 kg.

Calculate the weight of the oil.

weight of oil =

.....
.....
.....
(3 marks)

(d) The volume of the oil in the bucket is 0.0163 m^3 .

Calculate the density of the oil.

density =

.....
.....
(2 marks)

4 (a) An electric current transfers energy from the battery to the filament lamp.

State the two energy transfer pathways for the energy emitted by the filament lamp.

.....

.....
(2 marks)

(b) State which store of energy in the battery is decreasing.

.....
(1 mark)

(c) Explain how the principle of conservation of energy applies to this circuit.

.....
(1 mark)

5 (a) Complete the sentence below about energy sources.

Two methods of using energy from water are _____ and _____.

(2 marks)

(b) State two advantages of one of the methods from part (a).

(2 marks)

(c) A graph of the energy output of two renewable resources over time is shown in Fig. 1.1.
One is a solar cell, the other is a wind turbine.

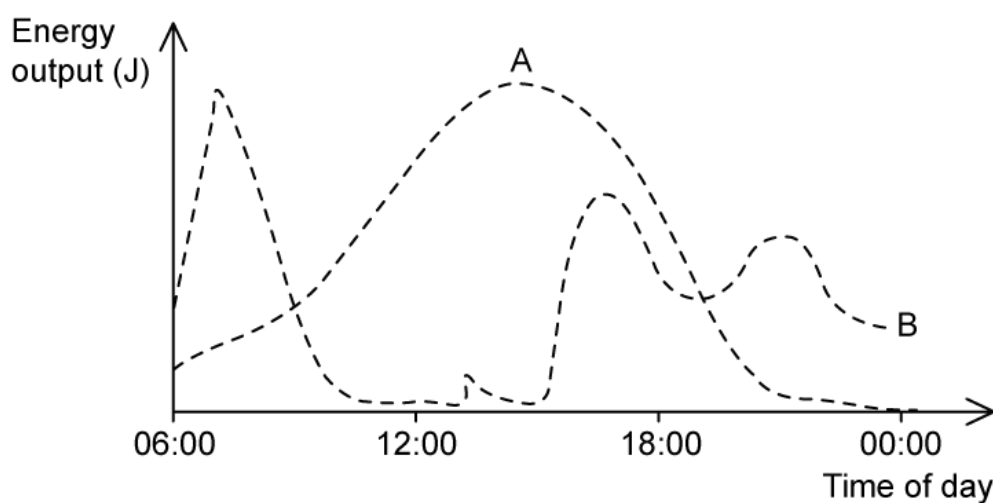


Fig. 1.1

Fig. 1.1

State which line on the graph represents the energy output of a solar cell.

(1 mark)

(d) Explain how you arrived at your answer to part (c).

.....

.....

(2 marks)

- 6 (a)** A student constructs a device for absorbing thermal energy from the Sun. Fig. 6.1 shows the device.

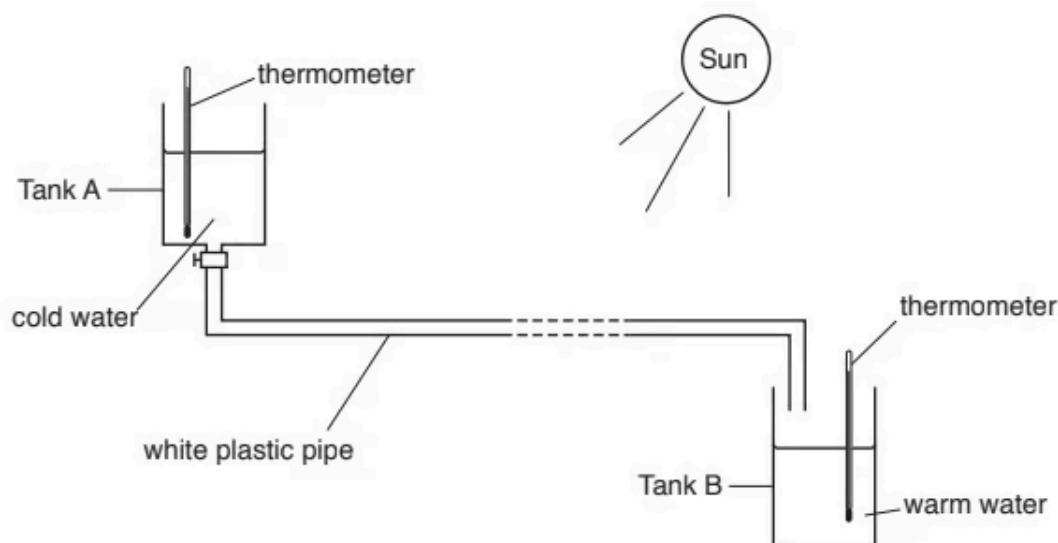
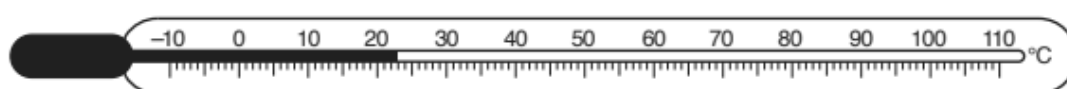


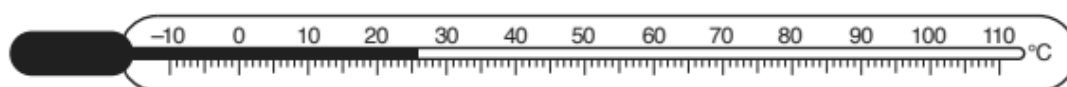
Fig. 6.1

The student places the white plastic pipe in sunlight. The cold water flows slowly from Tank A to Tank B. Energy from the Sun heats the water in the pipe.

Fig. 6.2 shows the temperatures in Tank A and Tank B.



Thermometer showing temperature in Tank A.



Thermometer showing temperature in Tank B.

Fig. 6.2

Determine the rise in temperature of the water.

temperature rise = °C

.....
(1 mark)

- (b)** The student wants to increase the thermal energy absorbed by the water in the pipe. Suggest three improvements he can make to increase the thermal energy absorbed.

.....
.....
.....
(3 marks)

- (c)** Describe how the thermal energy is transferred from the Sun to the water inside the pipe.

.....
.....
(2 marks)

7 (a) Circle two of the following that apply to an ultrasound wave travelling in air.

frequency 3.5 Hz

frequency 350 Hz

frequency 35 000 Hz

longitudinal

transverse

speed 1.5 m/s

speed 1.5×10^3 m/s

speed 1.5×10^6 m/s

.....
.....
(2 marks)

(b) Calculate the wavelength of X-rays with frequency 1.3×10^{17} Hz in a vacuum.

wavelength =

.....
.....
(3 marks)

(c) A dentist takes an X-ray photograph of a patient's teeth. Explain why it is safe for the patient to be close to the source of X-rays, but the dentist must stand away from the source.

.....
.....
(2 marks)

(d) State, with a reason, why microwave ovens are designed only to work with the door closed.

(2 marks)

8 (a) A student is given three unknown metal strips and a bar magnet.

The student is also given the following information.

1 strip is magnetic and magnetised

1 strip is magnetic but not magnetised

1 strip is not magnetic

Describe a method the student could use to identify each strip.

(4 marks)

(b) Describe the difference between a magnetised and an unmagnetised magnetic material.

(3 marks)

- 9 (a) Fig. 10.1 shows a circuit used by a student to test a metal wire made of nichrome.

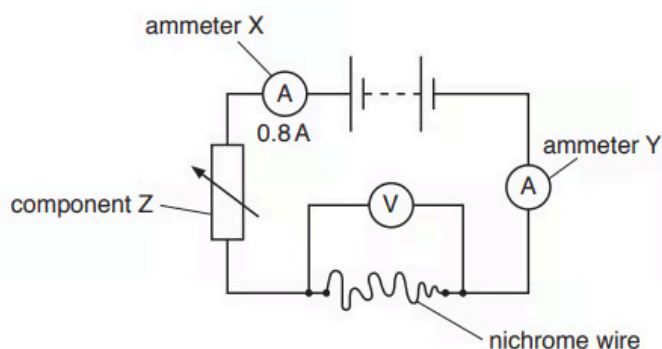


Fig. 10.1

State the name of component Z.

(1 mark)

- (b) The current reading on ammeter X is 0.8 A. State the reading on ammeter Y.

(1 mark)

- (c) The current in the nichrome wire is 0.8 A. The potential difference (p.d.) across the nichrome wire is 4.5 V.

Calculate the resistance of the nichrome wire.

resistance = Ω

(3 marks)

- (d) The student tests a different nichrome wire, which is thicker than the wire in (c), but of the same length. When testing this wire, the current in the wire is different from the value given in (c).

State and explain the difference in current.

(2 marks)

10 (a) A student does an experiment to determine the resistance of a fixed resistor, R .

The student draws an incomplete diagram of the circuit, as shown in Fig. 10.1.

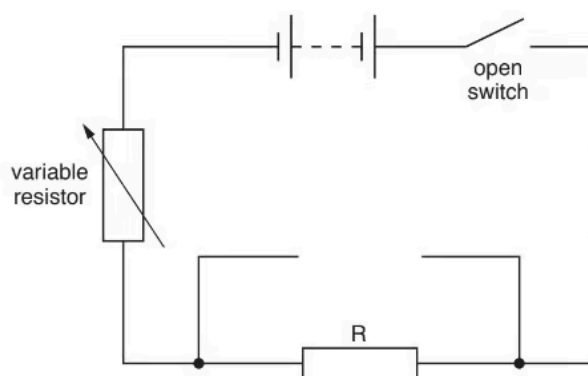


Fig. 10.1

(i) On Fig. 10.1, draw the missing circuit symbols.

[3]

(ii) Describe how the student could use the circuit to determine a reliable value for the resistance of R .

[5]

(8 marks)

(b) Fig. 10.2 shows a $20\ \Omega$ resistor connected to a power supply.

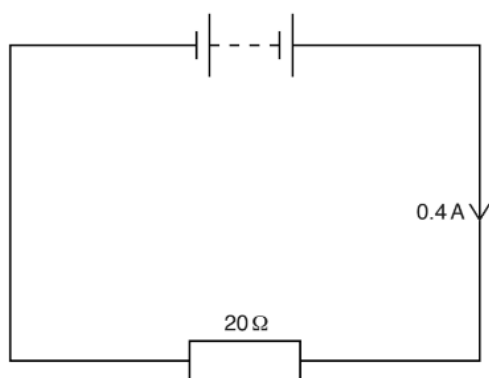


Fig. 10.2

A second $20\ \Omega$ resistor is connected in series with the first.

State and explain how this affects the current in the circuit.

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.....

.....

(4 marks)

- 11 (a)** The appearance of the Moon from Earth can be explained by the relative motion of the Moon and the Earth.

Explain why the Moon rises and sets.

.....
(1 mark)

- (b)** The Moon is said to undergo a lunar cycle.

(i) State how much time each lunar cycle takes to complete.

[1]

(ii) Explain why the Moon has phases which are seen throughout the lunar cycle.

[3]

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.....
.....
.....
(4 marks)

- (c)** Observers on Earth always see the same side of the Moon.

Explain why we never see the far side of the Moon when looking from Earth.

.....
.....
(2 marks)